# Cascadia Classifier Model Retraining Pipeline

## Use Case 1 – Model Maintainer

The Cascadia Classifier UI has a Feedback tab, intended to help identify misclassified images and images not present in the dataset. The general process for integrating the feedback into a retrained model is as follows: gather the landmarks specified in the feedback; add the landmarks to the landmarks_washington_full.csv dataset; put the landmark images into a folder, named after the landmark; add the folder to data/images; and re-run the model training script.

1. Gather the landmarks specified in the feedback. Each feedback form is emailed to the Cascadia Classifier email address.

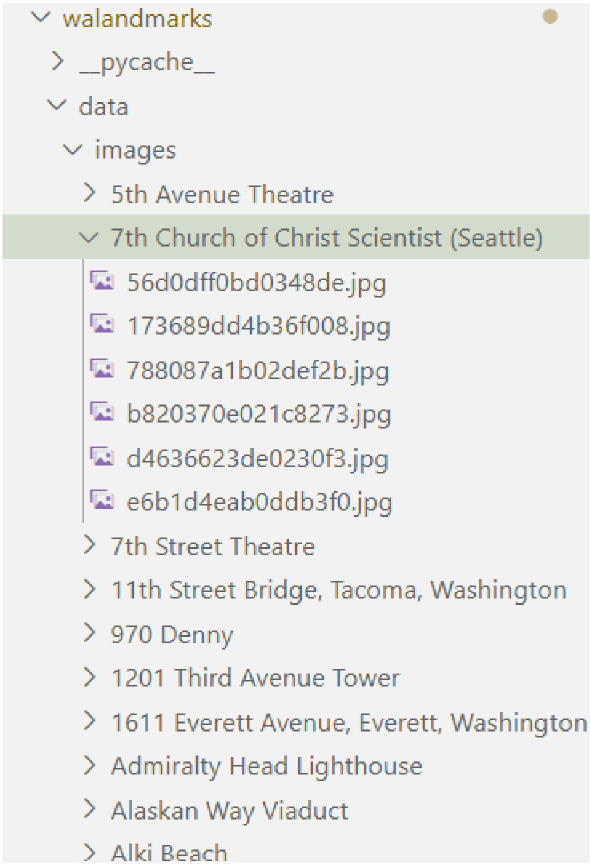
| | | |
|---|---|---|
| **me** | **New message from** - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann... | **2:00 PM** |
| **me** | **New message from** - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann... | **1:58 PM** |
| **me** | **New message from** - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann... | **1:55 PM** |
| **me** | **New message from** - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann... | **1:22 PM** |
| **me** | **New message from** - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann... | **1:11 PM** |
| me | New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni... | 12:57 PM |
| me | New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni... | 12:23 PM |
| me | New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Dem... | 12:15 PM |
| me | New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Izzy ... | 11:07 AM |
| me | New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni... | 10:23 AM |

2. Add the landmarks to the landmarks_washington_full.csv dataset. If the landmark is new to the dataset, assign it a new landmark_id.

```
landmark_id,name,supercategory,location,latitude,longitude,category
161,Lime Kiln Point State Park,Washington state park,"San Juan Island, San Juan County,
239,Illahee State Park,Washington state park,"Kitsap County, Washington, Pacific Northwe
262,Comet Falls,waterfall,"Mount Rainier National Park, Lewis County, Washington, Pacifi
291,"St. Andrews Episcopal Church (Chelan, Washington)",church building,"Washington, Pac
517,Titlow Beach,landform,"Tacoma, Pierce County, Washington, Pacific Northwest, Washing
763,St. Demetrios Greek Orthodox Church (Seattle),church building,"Washington, Pacific N
769,Beckler Peak,mountain,"King County, Washington, United States",47.736497222222226,-1
798,"Granite Mountain (King County, Washington)",mountain,"King County, Washington, Unit
1139,Lake Whatcom,lake,"Washington, Whatcom County, United States",48.73305555555556,-12
1255,"Bryant, Seattle, Washington",neighborhood,"Seattle, King County, Washington, Pacif
1528,Dome Peak,mountain,"Washington, Pacific Northwest, Washington, Pacific States Regio
1554,Potholes Reservoir,reservoir,"Washington, Pacific Northwest, Washington, Pacific St
1633,"Lincoln Theatre (Mount Vernon, Washington)",movie theater,"Mount Vernon, Skagit Co
1711,"Broadmoor, Seattle, Washington",neighborhood,"Washington, Pacific Northwest, Washi
2393,Steptoe Battlefield State Park,Washington state park,"Rosalia, Whitman County, Wash
```

3. Put the landmark images into a folder, named after the landmark.

4. Add the folder to data/images. The final structure after steps (3) and (4) is shown below.




5. Re-run the model training script. The model training script is located at walandmarks/notebooks/Landmark Classification EfficientNetB0.ipynb and will integrate the new data automatically if steps (1) – (4) are correct.